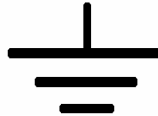




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Plasma System Earth Grounding Procedure Instructions



Prior to Ez-Plasma's technicians arrival at the cutting machine installation site, the following grounding procedure and test must be preformed. This grounding procedure will differ from location to location due to differing soil compositions. If the recommended earth ground of $\leq 3.00v$ cannot be attained, ground enhancing material may be required.

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Required materials:

Minimum of 1 - 12ft. 3/4” copper ground rod.
100 W 115VAC Incandescent (bulb type, NOT florescent)
Digital Voltmeter capable of measuring AC voltage
6ft. 12ga. Extension cord
1 AWG stranded wire or greater
3/4” Ground rod clamps

Grounding Procedure:

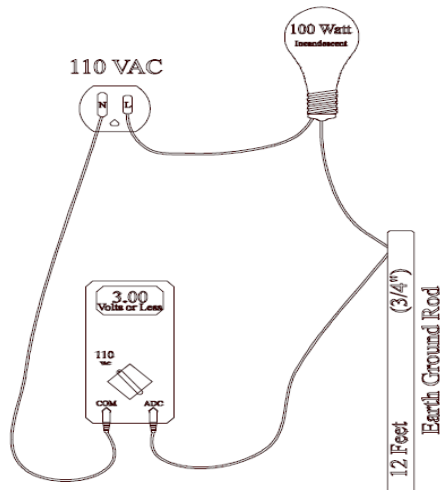
1. Locate the first ground-rod as close to the intended machine location as possible. Whether this be drilling a hole through the concrete slab in the facility or driving the rod into the soil directly outside the facility.
2. If necessary locate more holes with a minimum of 18ft. between the holes, since the grounds rods cannot be within 18ft. of each other.
3. If multiple ground rods are required, connect the ground rods using a 1/0 AWG wire or greater.
4. Test the ground rod system voltage using the testing procedure shown below.

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Grounding Test



System grounding check procedure:

1. Make sure no other cables are connected to the ground rod. If several rods are connected together, leave them connected.
2. Set the digital volt meter to read AC voltage.
3. If the meter is not auto-ranging, set it to 200VAC.
4. Verify that the 115 VAC receptacle is LIVE by plugging the test light into the receptacle with the 100 watt bulb installed and turned on.

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5. Plug the extension cord into the wall outlet.
6. Referring to the figure above, insert one side of the test light plug into the “HOT” side of the extension cord receptacle. If the outlet the test light is plugged into is wired correctly, the “HOT” side will be the narrower slot.
7. Insert the RED meter lead into the “NEUTRAL” side of the extension cord receptacle. If the outlet the test light is plugged into is wired correctly, the “NEUTRAL” side will be the wider slot.
8. Touch the other blade of the test light plug to the ground rod.
9. If the breaker trips immediately, the circuit may be on a GFI type receptacle, which will not permit such a testing method.
10. With the bulb socket test probe still attached to the ground rod, connect the red meter lead to the ground rod and record the displayed value.
11. A reading of .500 millivolts (3.00 VAC) or less indicates an optimum ground.
12. Other grounding procedures follow if a value less than 3.00 volt cannot be reached.

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